THE FLAVOURINATOR

SFU

ENSC 105W/100W
LA 31 THA
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Problem & Solution

Problem: cooking can be messy, inaccurate, and wasteful, people with tremors can experience difficulties accurately measuring

Solution: automated device that dispenses quantitative amounts of spice

- Reduce human error in cooking
  - **Audience**: elderly or people with disabilities that have difficulties measuring

Functional Requirements & Parameters

Functional Requirements:

- Should dispense a specific amount in volume (in teaspoon increments)
- Should autonomously dispense spice after input.

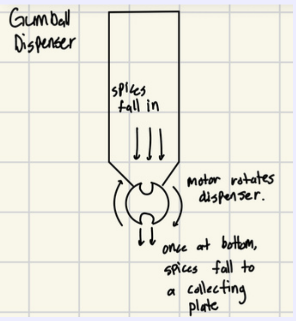
Parameters:

- Auger dimensions (height, scale, number of rotations)
- Funnel container (capacity, material)
- Stepper motor (angle, torque, speed)

Research

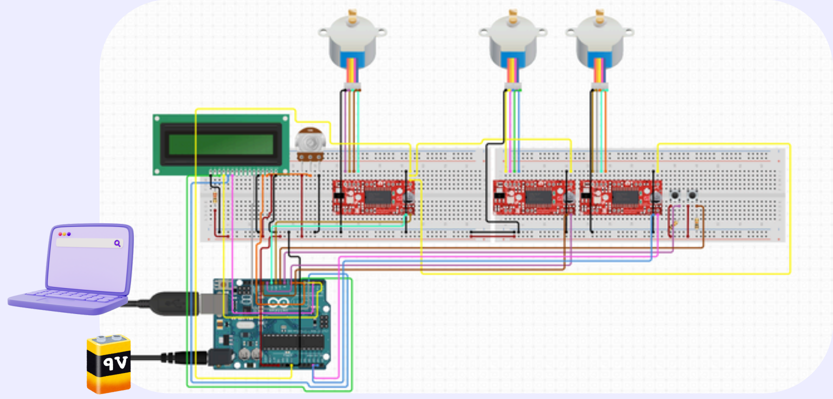
——————— Design Process ——————

- **Dispensing mechanism**
  - Initial approach - bubble gum dispenser
- **Motors + Drivers**
  - Initial approach - servo motors
    - Cannot rotate 360 degrees - cannot rotate our auger



Design

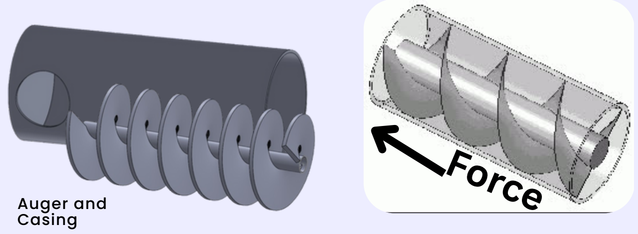
The Circuit:



Key Components

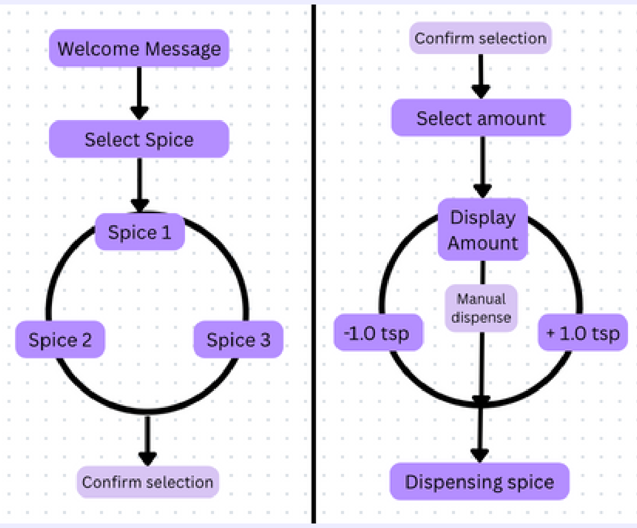
1) **LCD Interface: 2** toggle buttons for selection 16x2 LCD display
2) **Arduino Uno:** Transmits input signal to Drivers
3) **Stepper Motor:** Rotates Auger incrementally based on inputs

Auger Mechanism:

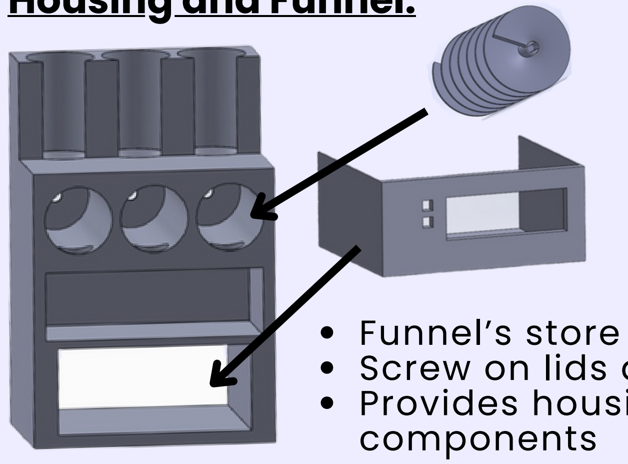


- Rotated incrementally by DC Stepper Motor
- Gravity and rotations propel spice down dispenser
- Increments produce precision up to ½ tsp

Software:



Housing and Funnel:



- Funnel's store spices and guide into auger
- Screw on lids allow spice refill
- Provides housing for interface, hiding circuit components

Outcomes

- **Dispense Accuracy:** ± 0.5 teaspoon of spice dispensed per operation (calibrated via 25 test trials each)
- **Dispense Speed:** completes dispensing ~6 seconds per rotation (faster than a human)
- **System Response Time:** < 1 second between user input and start of dispensing
- **Reliability: 20** consecutive each accurate dispenses without failure and resets when container is refilled

Improvements/Future Work

- Multiple Dispensing Containers: Expand system to hold more spices, allowing automated switches between them
- User Interface: Upgrade to a larger LCD or touchscreen for easier navigation and more options
- Calibration System: Create a self-calibration feature so the device adjusts for different spice densities and textures (trying more coarse spices)

Components & Budget

Component	Price (Total~$50)
2x Arduino Uno	Borrowed
16x2 LCD Display	Borrowed
2x Tact Touch Push Buttons	$2.00
x Potentiometer	Borrowed
5x 5V Stepper Motor + ULN2003 Driver	$16.99 (Amazon)
PLA Housing	$0 (SFU Library)
Filament	$28 (Amazon)
9V batteries	$5 (Dollarama)

Challenges

- Making sure the auger fits with the stepper motor
- Activating different motors according to the selected spice
- Confirm spices - was unresponsive
- Spice can be coarse - auger might get stuck
- Dispensed amount must be reasonable accurate

Resources